

Clock Tree Synthesis (CTS) Debug in IC Compiler II

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Agenda

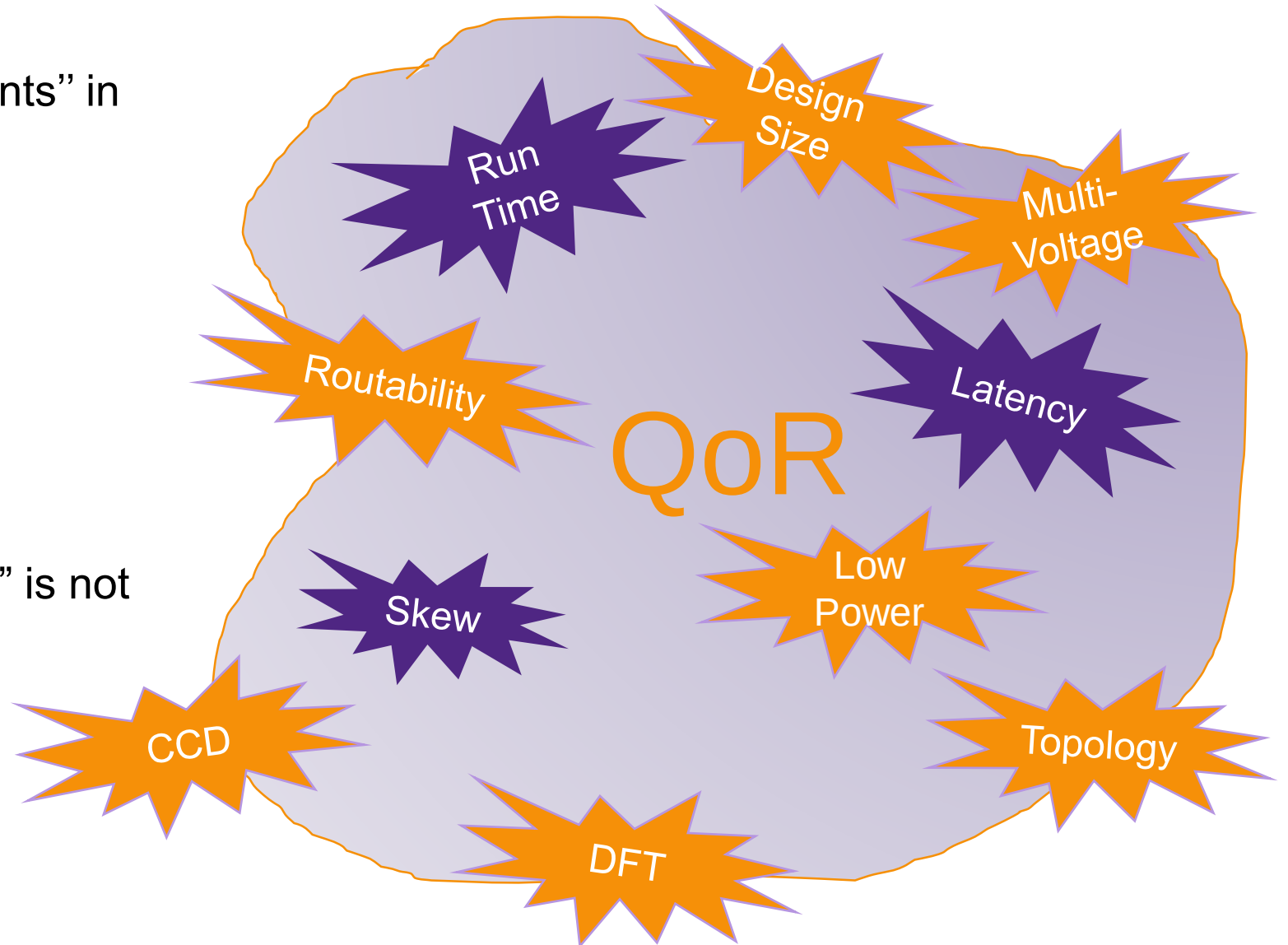
- **Introduction**
- Tools For Debug
- Appendix

Introduction

What are the typical “problem points” in CTS?

- Skew
- Latency
- Run Time
- Others ...

Very often, the “reported problem” is not the “actual problem”



Agenda

- Introduction
- **Tools For Debug**
 - Checks
 - Logs
 - Reports
 - GUI
 - Custom reports
- Appendix

check_clock_trees

Provides key information on CTS setup - All The Warnings In One Place

Summary			
Tag	Count	Solution	Description
Clock Definitions & Propagation			
CTS-019	0	None	Clocks propagate to output ports
CTS-907	24	None	There are disabled timing arcs in the clock network
Reference Cells			
CTS-008	0	None	Clock reference cells have dont touch or dont use
CTS-903	1327	None	Cells instantiated in the clock network are not in the clock reference list
Skew Balancing			
CTS-006	0	None	Balancing conflicts exist between different clocks
CTS-908	0	None	Large phase delay in abstracted sub-blocks
CTS-910	0	None	Balance point constraints are defined downstream of another balance point or ignore point constraint
CTS-911	0	None	Clock pins downstream of a balance point or ignore point have been added to a skew group
CTS-913	0	None	Explicit ignore points have been added to a skew group, and will not be balanced
CTS-917	0	None	Implicit ignore points have been added to a skew group, and will be balanced
Multi-Voltage			
CTS-901	0	None	Clock nets have MV violations
CTS-902	1	None	No AON (always-on) buffers or inverters available for CTS
Capacitance & Transition Constraints			
CTS-909	2	None	set load constraints detected in the clock tree
CTS-914	0	None	set_input_transition on clock ports exceeds the max transition limit
Other issues			
CTS-013	0	None	Cells in the clock network have a dont_touch constraint
CTS-015	24	None	set_max_delay or set_min_delay constraints are defined in the clock network

Organized in categories

"Details" Section at the end (not on the slide)

report_clock_settings

Provides key information on CTS setup - Configuration and Routing Rules

```
*****
Report : clock settings
*****
```

Configurations

```
=====
##Global
Mode = mode_norm.OC_F_0.RC_RCBEST
Corner = mode_norm.OC_F_0.RC_RCBEST
Max transition: 0.177
Max capacitance: 0.600
Target skew: 0 (default)
Target latency: Not specified
```

Global constraints
and targets: per
mode and
associated corners

```
Mode = mode_norm.OC_F_0.RC_RCBEST
##clk
Corner = mode_norm.OC_F_0.RC_RCBEST
...
```

Clock by clock constraints
and targets: per mode and
corner

Clock by clock
routing rule
settings

Routing Rules

```
=====
##Global
Routing rule for sink nets: icc2rm_leaf Min Layer:M2 Max Layer:M4
Routing rule for internal nets: icc2rm_2w2s_shield_list Min Layer:M5 Max Layer:M6
Routing rule for root nets: icc2rm_2w2s_shield_list Min Layer:M5 Max Layer:M6
Routing rule for all nets: Not specified
Net-based routing rules (by set_clock_routing_rule): Not specified
Net-based routing rules (by set_routing_rule): Not specified
```

report_clock_settings

Provides key information on CTS setup - Cells and Cell Spacing

```
*****
Report : clock settings
*****
```

... Buffers and Inverters

===== ##Buffers

Name	Area	Max Transition	Max Load
UTRDL/BUF_S_12	1.037	0.940	2.732

##Inverters

Name	Area	Max Transition	Max Load
UTRDL/INV_S_6	0.415	0.940	1.035
UTRDL/INV_S_8	0.518	0.940	1.375

...

Spacing Rules

===== ##By design

No by design spacing rule is specified

##By clock

No by clock spacing rule is specified

##By libcell

No by libcell spacing rule is specified

##By clock and libcell

No by clock and libcell spacing rule is specified

Full list of cells made
available for CTS use

Cell spacing settings by
design, clock and libcell

“Does all of
this match my
intentions?”



Tools For Debug

- Checks
- **Logs**
 - **clock_opt –to build_clock**
- Reports
- GUI
- Custom reports

Log File Information

Use the Headers

*** CTS STEP: Design Initialization**

Information: All clock objects will be converted from ideal to propagated clock during CTS. (CTS-105)

Information: CTS will work on the following scenarios. (CTS-101)

s1 (Mode: func1; Corner: func_BEST)

s2 (Mode: func1; Corner: func_WORST)

Information: CTS will work on all clocks in active scenarios, including 1 master clocks and 0 generated clocks. (CTS-107)

CTS related app options set by user:

cts.compile.enable_local_skew = true

Buffer/Inverter reference list for clock tree synthesis:

stdcell_lib/BUFX10

CTS NDR rule list:

Design Base; Net Type: root; Rule: 2w2s; Min Layer: M6; Max Layer: M7

Design Base; Net Type: internal; Rule: 2w2s; Min Layer: M6; Max Layer: M7

Design Base; Net Type: sink; Rule: 1w2s; Min Layer: M4; Max Layer: M7

*** CTS STEP: Existing Clock Tree Removal**

A total of 1 buffer(s) and 0 inverter(s) have been removed.

*** CTS STEP: Clock Tree Initialization**

Start Auto-Exception Derivations...

No internal pin was found.

No conflict pin was found.

No macro pin was found for clock balance point settings.

Structured log file where each step is separated by an easily searched header.

Details are provided within each step

Log File Information

Gate by Gate Synthesis - info details can be very useful

```
*****
* CTS STEP: Clock Cell Relocation
*****
Inst 'mux1' is not movable
Information: Relocated the clock cell 'icg3' from (791.37, 352.30) to (188.19, 87.10). (CTS-106)
Inst 'b2' is not movable
Information: Relocated the clock cell 'icg1' from (795.42, 744.90) to (989.96, 370.50). (CTS-106)
A total of 3 clock cells have been relocated
```

At each gate level, the details about driver name, clocks considered for synthesis, number of loads, drc constraint, and more are displayed

```
*****
* CTS STEP: Gate-By-Gate Clock Tree Synthesis
*****
```

```
Gate level 2 clock tree synthesis
driving pin = mux1/Z
```

```
Clocks:
```

```
clk (func1)
```

```
Design rule constraints:
```

```
max transition = 0.100000
```

```
max capacitance = 0.600000
```

```
Number of load sinks = 1
```

```
Number of ignore points = 0
```

```
Warning: Gate 'mux1' is not sizable because of lib_cell dont_touch. (CTS-041)
```

```
Information: Exception 'freeze port check' is applied on "xxx" (CTS-062)
```

```
.....
```

```
Added 5 Repeaters. Built 5 Repeater Levels
```

```
Phase delay: mux1/D0 : (lp max: 0.194 sp min: 0.194) : skew = 0.000
```

```
...
```

```
Information: The run time for gate-by-gate clock tree synthesis is 0 hr : 0 min : 1.35 sec, cpu time
```

```
*****
```

```
* CTS STEP: Latency Bottleneck Analysis
```

```
*****
```

Check for warnings

Phase delay and Skew are reported and will be used at higher gate levels later in the process

Show large jumps in latency at the end of the gate-by-gate

Log File Information

Constant QoR Updates, Throughout the Flow

```
*****
* CTS STEP: DRC Fixing Beyond Exceptions
*****
```

```
...
```

```
*****
* CTS STEP: Clock Net Global Routing
*****
```

```
Total number of global routed clock nets: 28
```

```
Information: The run time for clock net global routing is 0 hr : 0 min : 0.86 sec, cpu time is 0 hr : 0 min : 0.86 sec. (CTS-104)
```

```
...
```

```
*****
* CTS STEP: Pre-Optimization DRC Fixing
*****
```

```
Fixing clock: clk1 mode: func1 root: clk1
```

```
Information: CTS QoR Pre Initial DRC Fixing: GlobalSkew = 0.0068; ID = 0.4446; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000; ClockBufArea = 34.7490; ClockCellArea = 51.7725; Clock = clk; Mode = func1; Corner = func_BEST; ClockRoot = clk1. (CTS-037)
```

```
Information: CTS QoR Pre Initial DRC Fixing: GlobalSkew = 0.0096; ID = 0.4640; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000; ClockBufArea = 34.7490; ClockCellArea = 51.7725; Clock = clk; Mode = func1; Corner = func_WORST; ClockRoot = clk1. (CTS-037)
```

```
Starting fixing clock tree DRC Violation ...
```

```
Resized 0 cell(s), relocated 0 cell(s), cloned 0 cell(s) and inserted 0 buffer(s)/inverter(s)
```

```
Ran incremental ZGR ....
```

```
Information: CTS QoR Post Initial DRC Fixing: GlobalSkew = 0.0068; ID = 0.4446; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000; ClockBufArea = 34.7490; ClockCellArea = 51.7725; Clock = clk; Mode = func1; Corner = func_BEST; ClockRoot = clk1. (CTS-037)
```

```
Information: CTS QoR Post Initial DRC Fixing: GlobalSkew = 0.0096; ID = 0.4640; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000; ClockBufArea = 34.7490; ClockCellArea = 51.7725; Clock = clk; Mode = func1; Corner = func_WORST; ClockRoot = clk1. (CTS-037)
```

```
...
```

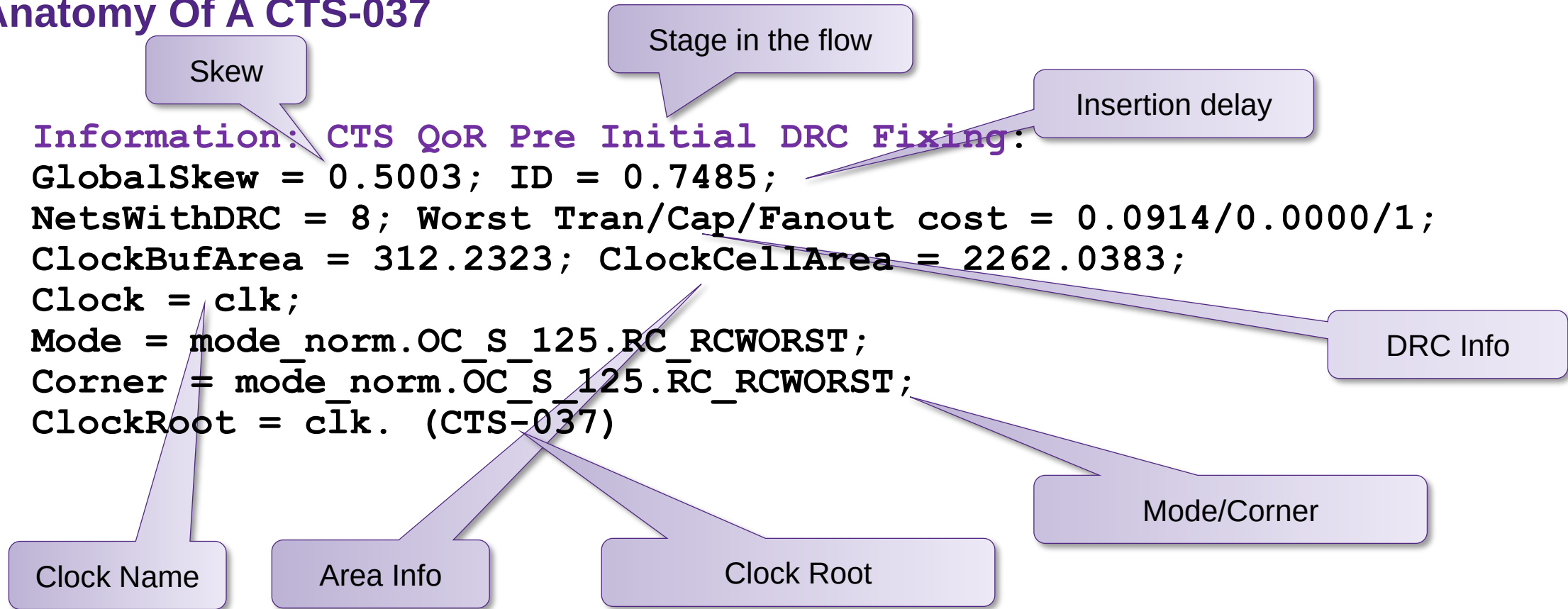
```
Information: The run time for pre-optimization DRC fixing is 0 hr : 0 min : 0.00 sec, cpu time is 0 hr : 0 min : 0.00 sec. (CTS-104)
```

QoR information for each clock, corner, and mode at the start and end of each optimization step. See the CTS-037 message for details.

See the CTS-104 message to get the runtime information for each step

Log File Information

Anatomy Of A CTS-037



You may need to filter the log to get some initial info. For example:

```
grep "CTS-037" logfiles/clock_opt_cts.log | grep "Clock = clk" | grep OC_S_125.RC_RCWORST
```

Log File Information

```
*****
* CTS STEP: Skew Latency Optimization and Area Recovery
*****
```

Optimizing clock tree

```
clock: clk mode: func1 root: clk1
clock: clk2 mode: func1 root: clk2
```

List clocks which are optimized together

```
Information: CTS QoR Pre Optimization: GlobalSkew = 0.0068; ID = 0.4446; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000;
ClockBufArea = 34.7490; ClockCellArea = 51.7725; Clock = clk; Mode = func1; Corner = func_BEST; ClockRoot = clk1. (CTS-037)
Information: CTS QoR Pre Optimization: GlobalSkew = 0.0096; ID = 0.4640; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000;
ClockBufArea = 34.7490; ClockCellArea = 51.7725; Clock = clk; Mode = func1; Corner = func_WORST; ClockRoot = clk1. (CTS-037)
```

QoR before optimization per clock and corner

Begin Network Flow Based Optimization:

```
Default network flow optimizer made 27 successful improvements out of 58 iterations
Resized 4, relocated 12, deleted 4, inserted 0, sizeUp Relocated 0 cells
```

```
Information: CTS QoR Post Optimization: GlobalSkew = 0.0064; ID = 0.3539; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000;
ClockBufArea = 30.8880; ClockCellArea = 47.9115; Clock = clk; Mode = func1; Corner = func_BEST; ClockRoot = clk1. (CTS-037)
Information: CTS QoR Post Optimization: GlobalSkew = 0.0024; ID = 0.3650; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000;
ClockBufArea = 30.8880; ClockCellArea = 47.9115; Clock = clk; Mode = func1; Corner = func_WORST; ClockRoot = clk1. (CTS-037)
```

Begin Area Recovery Buffer Removal:

```
AR: deleted 0 cell(s)
```

```
Information: CTS QoR Post Area Recovery: GlobalSkew = 0.0064; ID = 0.3539; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000;
ClockBufArea = 30.8880; ClockCellArea = 47.9115; Clock = clk; Mode = func1; Corner = func_BEST; ClockRoot = clk1. (CTS-037)
Information: CTS QoR Post Area Recovery: GlobalSkew = 0.0024; ID = 0.3650; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000;
ClockBufArea = 30.8880; ClockCellArea = 47.9115; Clock = clk; Mode = func1; Corner = func_WORST; ClockRoot = clk1. (CTS-037)
```

QoR after optimization, look for longest/shortest paths, large skew, nets with DRC on a clock and corner basis

Begin Area Recovery Resizing:

```
AR: resized 0 out of 24 cell(s)
```

```
Information: CTS QoR Post Area Recovery: GlobalSkew = 0.0064; ID = 0.3539; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000;
ClockBufArea = 30.8880; ClockCellArea = 47.9115; Clock = clk; Mode = func1; Corner = func_BEST; ClockRoot = clk1. (CTS-037)
Information: CTS QoR Post Area Recovery: GlobalSkew = 0.0024; ID = 0.3650; NetsWithDRC = 0; Worst Tran/Cap cost = 0.0000/0.0000;
ClockBufArea = 30.8880; ClockCellArea = 47.9115; Clock = clk; Mode = func1; Corner = func_WORST; ClockRoot = clk1. (CTS-037)
```

Log File Information

Summary Report – This May Be The Place To Start

```
*****
* CTS STEP: Postlude
*****
Marking clock synthesized attributes
Successfully legalize placement.
*****
* CTS STEP: Summary report
*****
There are 2093 flat clock tree nets.
There are 2102 non-sink instances (total area 2113.57) on clock trees including 0 instances dont_touch.
Clock tree synthesize and optimization added 231 buffers and 534 inverters (total area 390.15).
52 buffers/inverters were inserted below 30 leaf level Gates.
Compilation of clock trees finished successfully
Run time for cts 01:04:40.42u 00:05:46.21s 00:38:48.58e:

Skew Bottleneck Analysis:
Largest skew jumps (> 0.1 or 50 percent of the global skew) of the terms for clock clk:
Skewgroup: default_clk, Corner: func_BEST
Skew jumped by 0.010 at term icg2/GCP

Summary of messages during CTS:
=====
Tag  Count  Type      Description
-----
CTS-038  1 Warning No clock routing rule is specified.
CTS-041  1 Warning Gate '%s' is not sizable because of %s dont_touch.
CTS-043  1 Warning Gate '%s' is not sizable because of fixed placement
```

Check for large displacements

Details of the final synthesis result

Bottleneck Analysis

Look for a message summary to see what issues and notes have been identified by the tool during the synthesis process

Log file verbosity can be controlled, make sure it's on (the default)

```
icc2_shell> set_app_options -list {cts.common.verbose 1}
```

Tools For Debug

- Checks
- Logs
- **Reports**
 - report_clock_qor
- GUI
- Custom reports

report_clock_qor -type

Reports to Analyze the CTS QoR

- To report the clock tree QoR, use the report_clock_qor command
- It generates two types of report
 - Tabular reports
 - Histogram report

Report Type	Metrics
Tabular Reports (-type)	summary (latency, skew, area, and DRC)
	structure
	latency
	drc_violators
	robustness
	area
	balance_group
	local_skew
	power

Run also Pre-CTS and keep it

Check for largest and shorts paths to judge on skew

report_clock_qor -type

Histogram Options

Report Type	Metrics
Histogram Reports (-histogram_type)	latency
	level
	robustness
	wire_delay_fraction
	fanout
	capacitance
	transition
	wire_delay
	cell_delay
	local_skew
	wire_arc_length
	wirelength

See the report_clock_qor man page for full list of options

report_clock_qor –type summary

Organized and Structured Reporting

Attributes
=====

M Master Clock
G Generated Clock
U User Defined Skew Group
D Default Skew Group
* Generated Clock Balanced Separately

The clocks are organized by modes within the table

One table printed per corner

Summary Table for Corner s1										
Clock / Skew Group	Attrs	Sinks	Levels	Buffer Count	Buffer Area	Total Area	Max Latency	Global Skew	Trans DRC Count	Cap DRC Count
### Mode: s1, Scenario: sc1.BC clk4	M	96	5	0	0.000	25.688	--	--	0	0
default_clk4	D	92	--	--	--	--	1.045	1.996	--	--
gen2	G	60	3	0	0.000	7.774	1.036	1.996	0	0
gen4	G	49	1	0	0.000	0.000	1.002	1.996	0	0
sg1	U	4	--	--	--	--	0.048	0.004	--	--
### Mode: s2, Scenario: s2::s1 clk3	M,D	96	5	0	0.000	25.688	0.048	0.031	0	0
All Clocks		96	5	0	0.000	25.688	1.045	1.996	0	0

The “Attrs” column show master clock, generated clock, and skew group relationships within a scenario

Corner s2						
Clock / Skew Group	Buffer Area	Total Area	Max Latency	Global Skew	Trans DRC Count	Cap DRC Count
### Mode: s2, Scenario: sc1.w clk3	0.000	25.688	0.053	0.034	0	0
All Clocks	0.000	25.688	0.053	0.034	0	0

report_clock_qor -show_verbose_paths

Added Verbosity in Path Report

Point	Fanout	Cap	Trans	Incr	Path	Location	NDR	Layers	CTS Tags
source latency				0.0000	0.0000				
clk1 (in)	1	0.0258	0.0000	0.0000	0.0000 r	(999.92,737.10)	2w2s		clk
cts_inv_34763893/A (INVX8)			0.0327	0.0166	0.0166 r	(825.66,625.11)		{M7 113} {M6 174}	
cts_inv_34763893/Z (INVX8)	2	0.0223	0.0210	0.0138	0.0304 f	(825.72,624.72)	2w2s	{M7 113} {M6 174}	
...									
cts_inv_34683885/A (INVX8)			0.0549	0.0226	0.1336 f	(310.64,1.11)		{M6 171} {M7 287}	
cts_inv_34683885/Z (INVX8)	2	0.0215	0.0242	0.0177	0.1512 r	(310.70,0.71)	2w2s	{M6 300} {M7 10}	
mux1/D1 (CLKMUX2X2)			0.0485	0.0196	0.1708 r	(10.80,10.98)		{M6 300} {M7 10}	
mux1/Z (CLKMUX2X2)	1	0.0157	0.0337	0.0491	0.2199 r	(11.88,10.85)	2w2s		
								{M6 202}	dt_lib
cts_inv_33553772/A (INVX4)			0.0413	0.0111	0.2309 r	(214.38,16.24)		{M6 202}	
cts_inv_33553772/Z (INVX4)	1	0.0149	0.0268	0.0180	0.2489 f	(214.48,16.26)	2w2s		
...									
cts_inv_33523769/A (INVX8)			0.0305	0.0093	0.3169 f	(557.68,381.09)		{M6 46} {M7 230}	
cts_inv_33523769/Z (INVX8)	1	0.0161	0.0165	0.0131	0.3300 r	(557.75,381.48)	1w2s		
								{M5 229} {M4 47}	
sink4/CP (DFF1)			0.0299	0.0117	0.3418 r	(605.61,610.54)		{M5 229} {M4 47}	sink
total clock latency					0.3418				

Extra columns in clock path reporting for NDRs, routing layers, location, and so on provides detailed information and is helpful for debugging

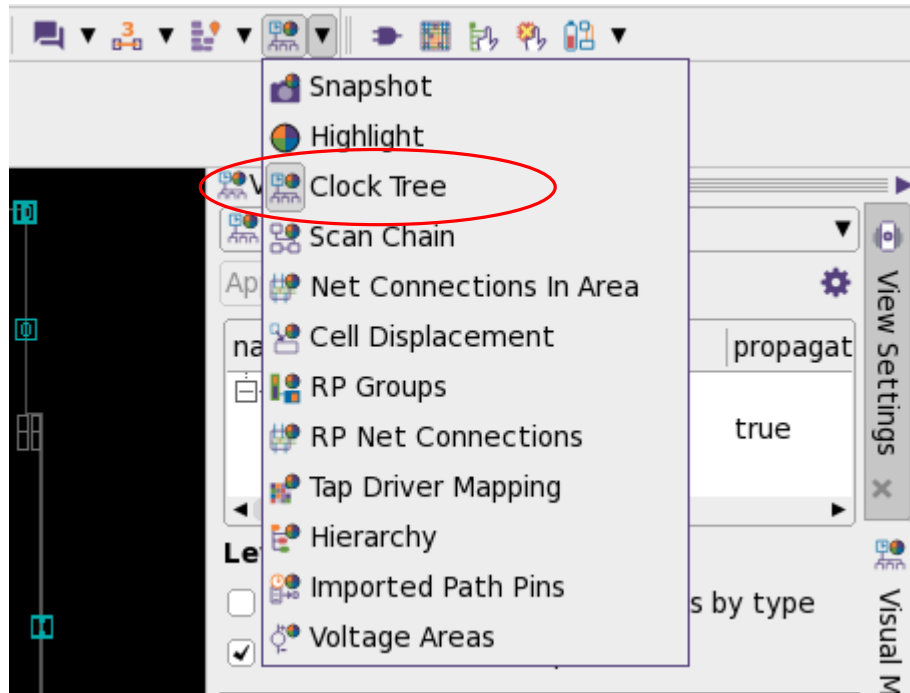
Tools For Debug

- Checks
- Logs
- Reports
- **GUI**
 - Clock tree Visual mode
 - Clock tree Analyzer
- Custom reports

Clock Visual Mode

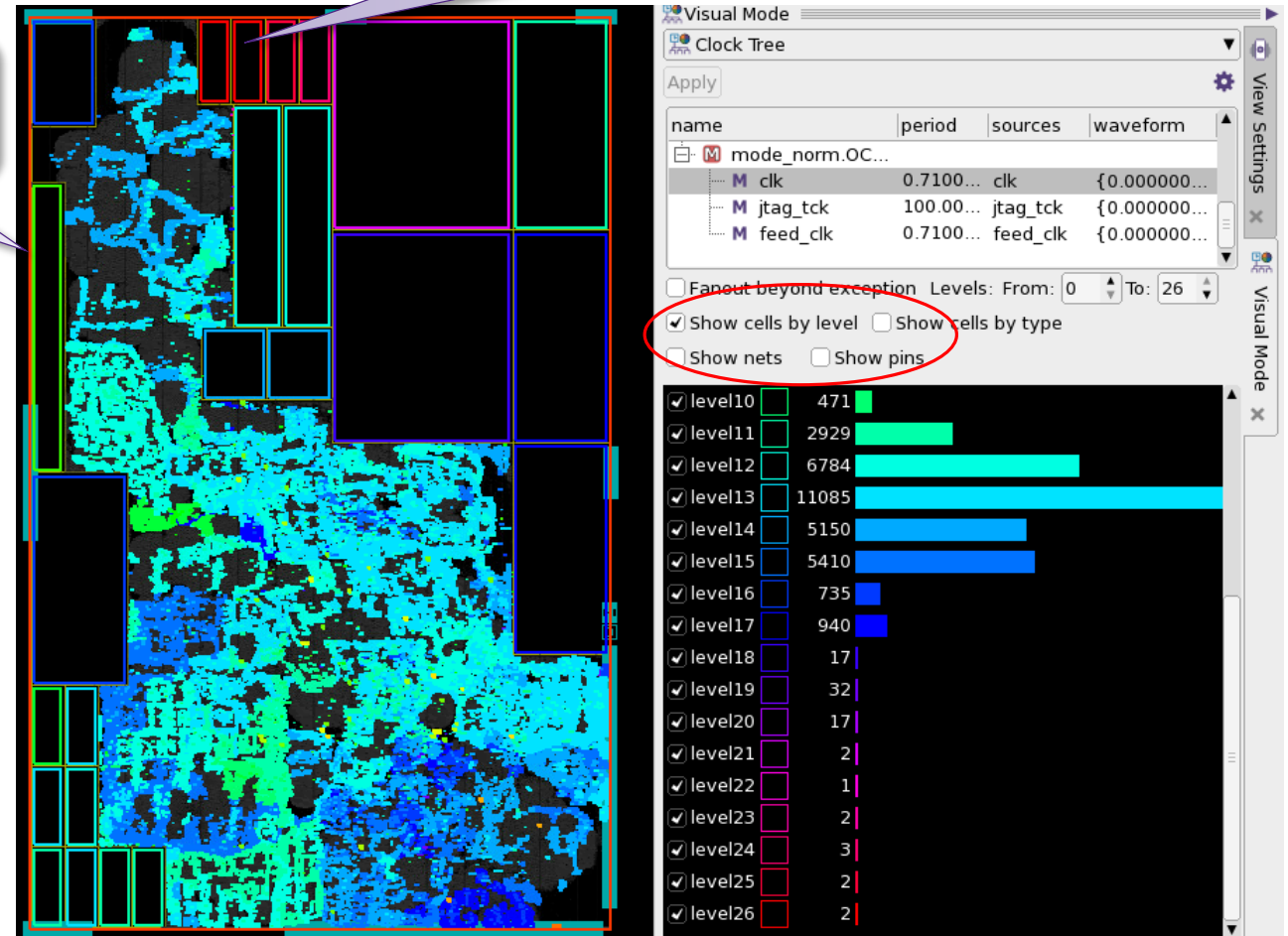
See tree cell and route topology

- Once the form is up, select a clock from the top scroll list.



Much earlier connection in the tree

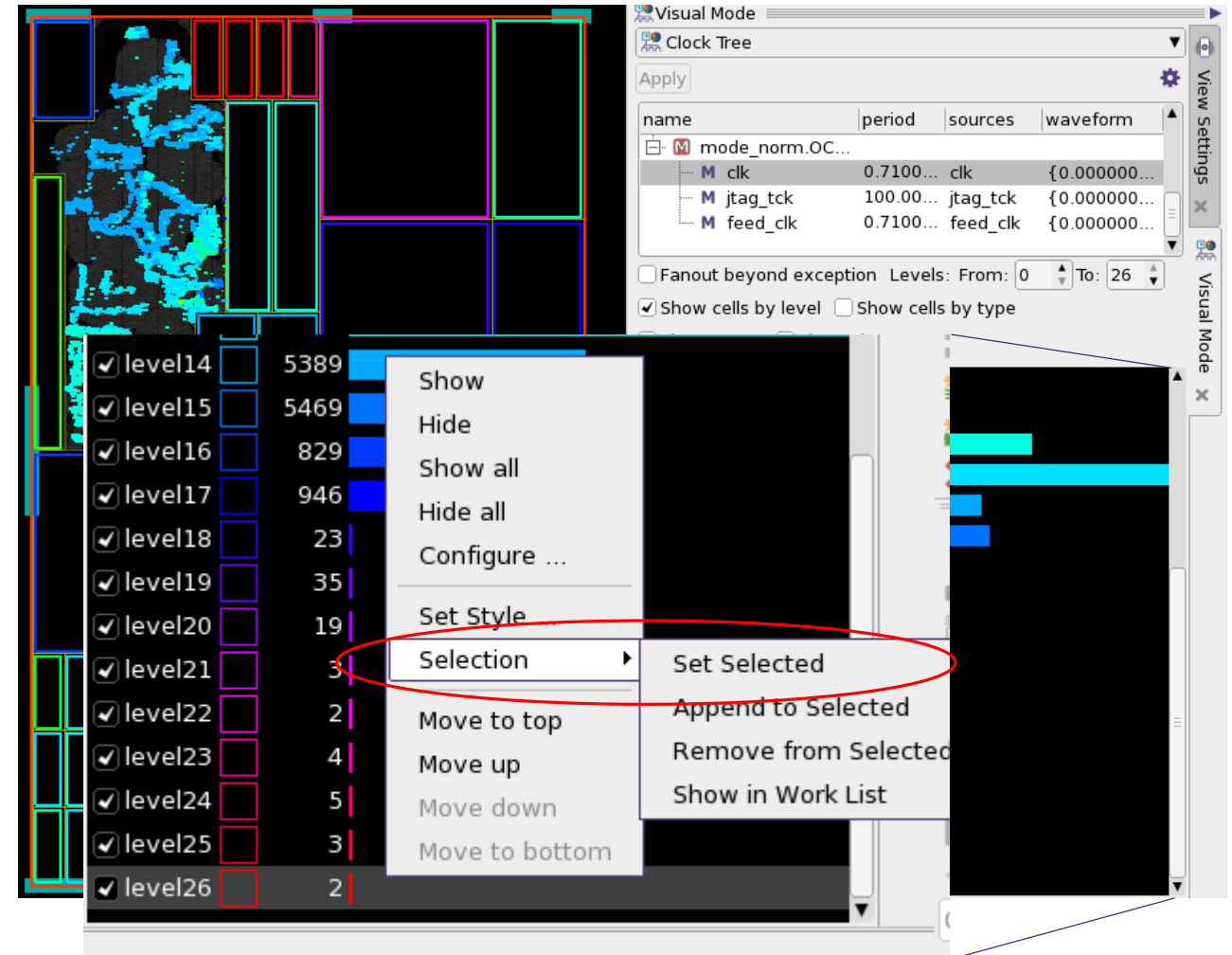
Latest connections in the tree



Clock Visual Mode

See tree cell and route topology

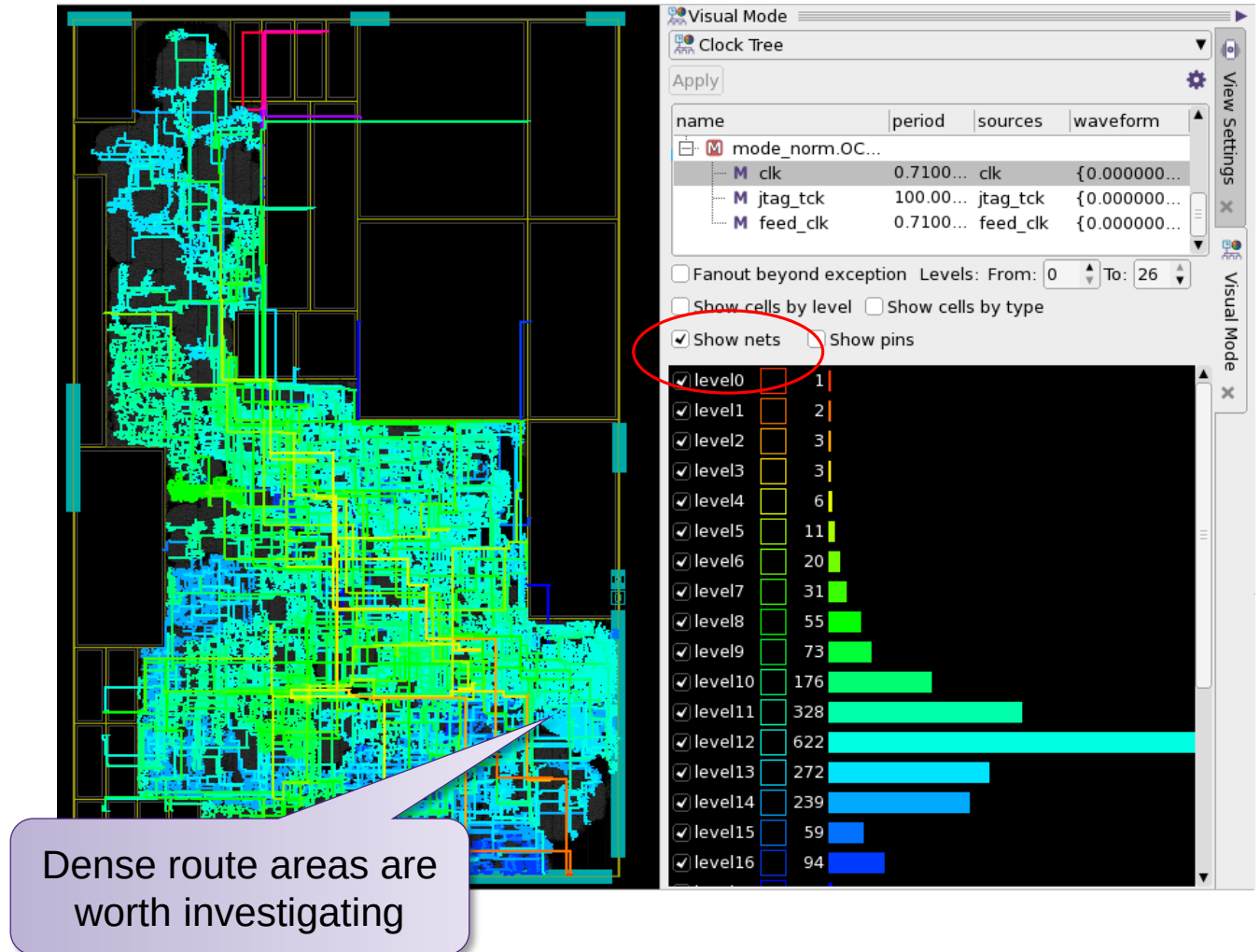
- Context sensitive menu provides some additional short cuts
 - Showing/Hiding rows
 - Changing global selection
 - Closer investigation
 - Cross probe in display or browser



Clock Visual Mode

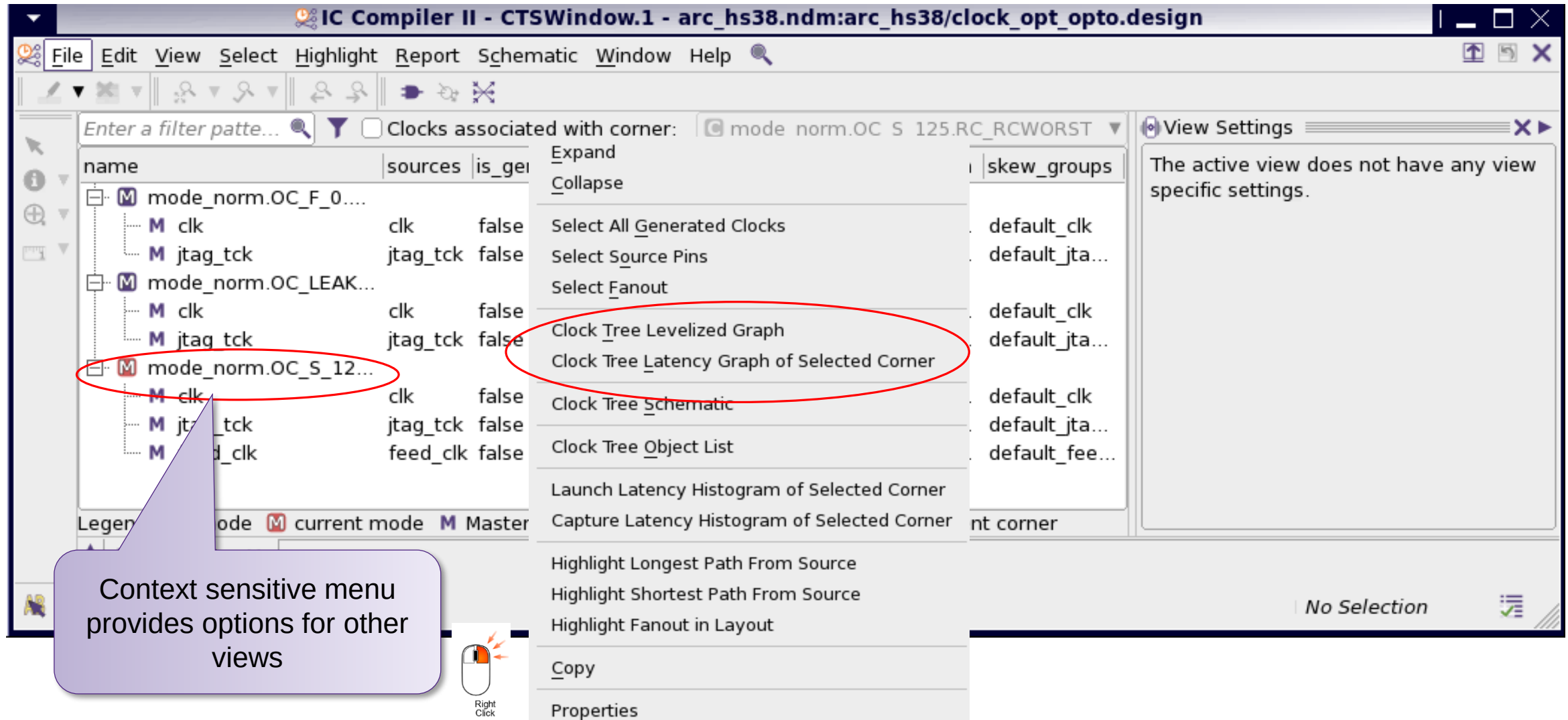
See tree cell and route topology

- Viewing nets gives route topology insight
- Cross probe in display or browser



Clock Tree Analysis Window

Clock Tree Analysis Window



The screenshot shows the IC Compiler II interface with the Clock Tree Analysis Window open. The window title is "IC Compiler II - CTSWindow.1 - arc_hs38.ndm:arc_hs38/clock_opt_opto.design". The menu bar includes File, Edit, View, Select, Highlight, Report, Schematic, Window, and Help. The toolbar contains various icons for navigation and analysis. The main area displays a tree view of clock nodes. The node "mode_norm.OC_S_125.RC_RCWORST" is selected and highlighted with a red circle. A context menu is open over this node, listing various actions. The menu items are:

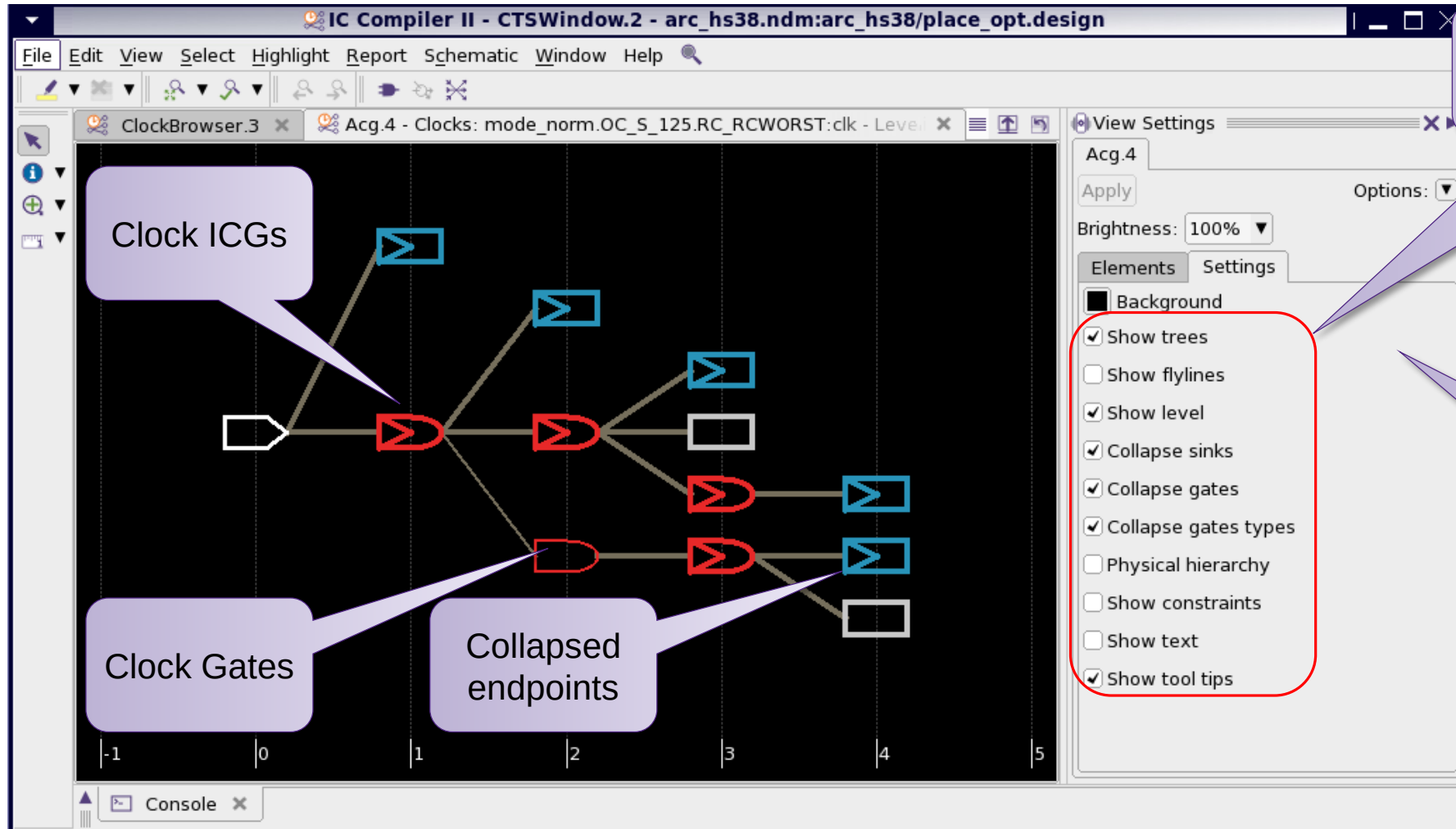
- Expand
- Collapse
- Select All Generated Clocks
- Select Source Pins
- Select Fanout
- Clock Tree Levelized Graph
- Clock Tree Latency Graph of Selected Corner
- Clock Tree Schematic
- Clock Tree Object List
- Launch Latency Histogram of Selected Corner
- Capture Latency Histogram of Selected Corner
- Highlight Longest Path From Source
- Highlight Shortest Path From Source
- Highlight Fanout in Layout
- Copy
- Properties

A callout box with a mouse icon and the text "Right Click" points to the context menu. A purple speech bubble contains the text: "Context sensitive menu provides options for other views".

On the right side of the window, there is a "View Settings" panel with a message: "The active view does not have any view specific settings." The status bar at the bottom right shows "No Selection".

Clock Tree Analysis Window

Clock Tree Analysis Window – Levelized Graph

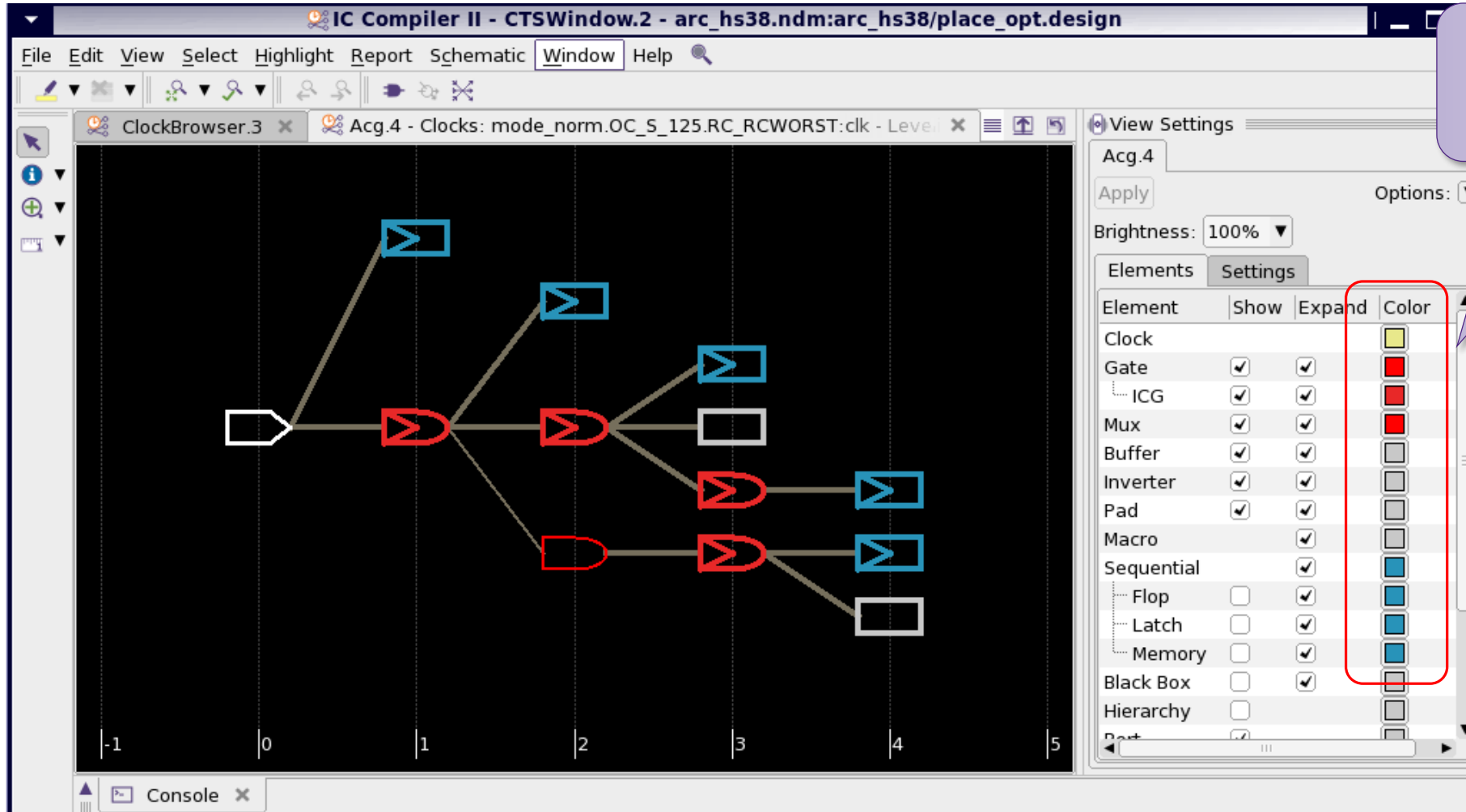


Settings adjustments may help simplify the graph

Look at this before CTS

Clock Tree Analysis Window

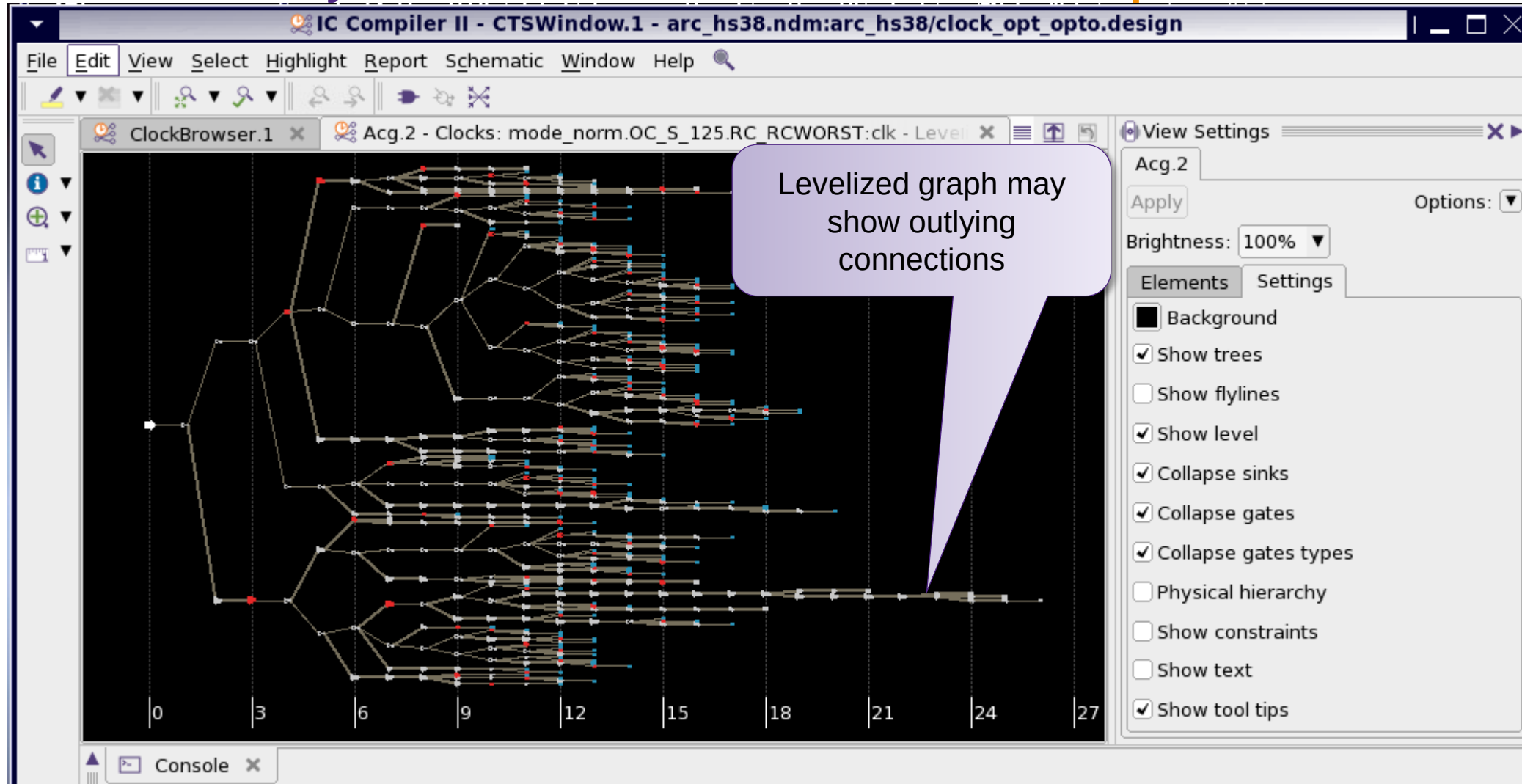
Clock Tree Analysis Window – Levelized Graph



Change the element colors to better suit your needs

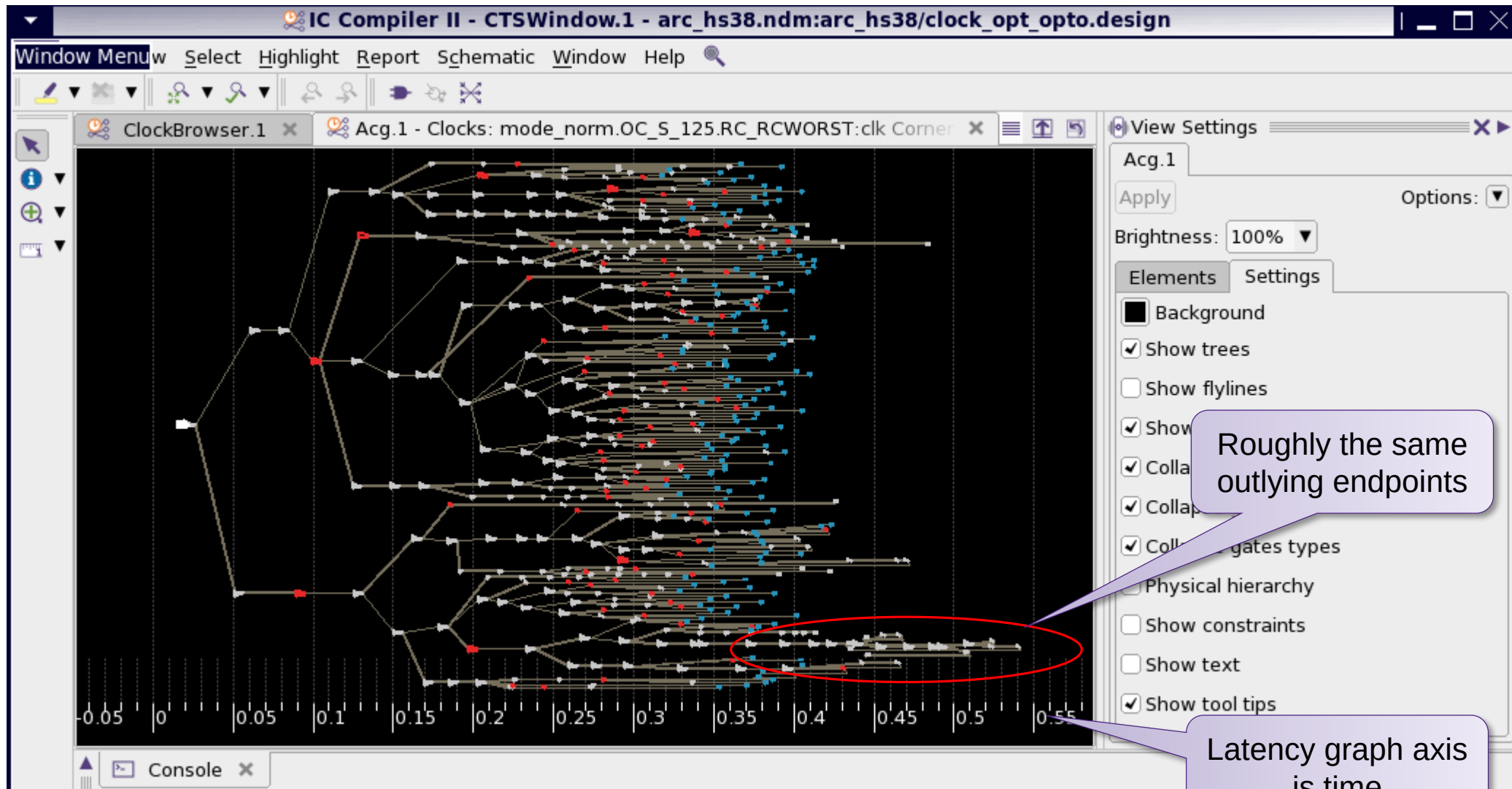
Clock Tree Analysis Window

Clock Tree Analysis Window – Post CTS Levelized Graph



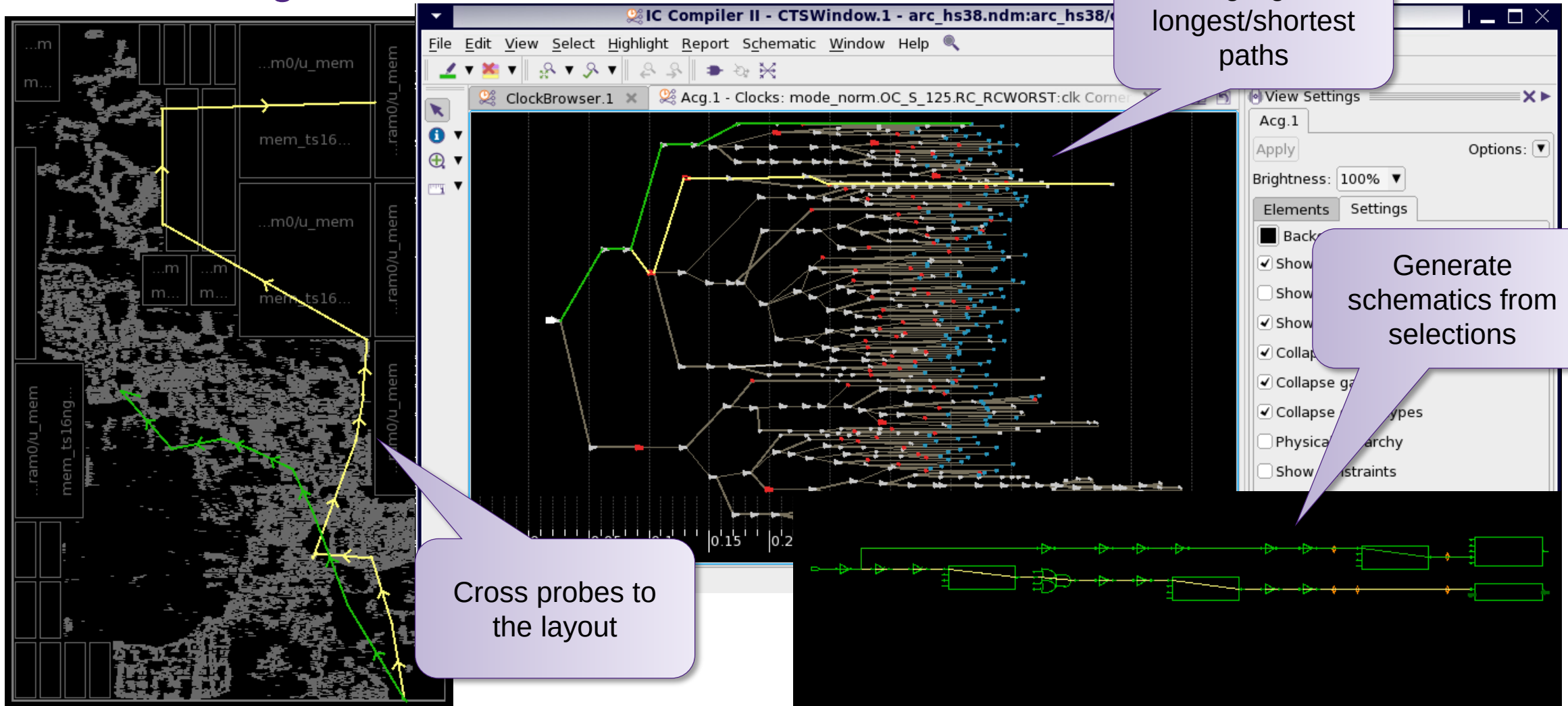
Clock Tree Analysis Window

Clock Tree Analysis Window – Post CTS Latency Graph



Clock Tree Analysis Window

Skew and Longest/Shortest Path



Tools For Debug

- Checks
- Logs
- Reports
- GUI
- **Custom reports**
 - `get_clock_tree_pins`
 - Clock related attributes

get_clock_tree_pins

- `get_clock_tree_pins` **returns a collection of clock tree pins**
 - The ability to filter the pins, is the real power of the function
 - Loads of clock tree related attributes available for querying/filtering

```
get_clock_tree_pins # get clock tree pins by various criteria
[-clocks object_list] (List of clocks)
[-from list_of_pins] (include pins on paths starting from these pins)
[-to list_of_pins] (include pins on paths ending at these pins)
[-filter extended_attribute_expression]
                    (filter by this attribute)
[-sort_by extended_attribute_expression]
                    (sort by this attribute )
[-index_range start_index_end_index]
                    (return a subset of the sorted pins)
[-groups_from list_of_pins]
                    (return a list of collections, one for each pin listed here)
[-metrics expression_list]
                    (one or more expressions to evaluate for each pin selected )
[-assign_to_variable variable_name]
                    (put results in associative array variable)
```

Target clocks, or
parts of clock

Filter, sort and limit
selection based on
attributes

Return the pin
metrics for further
processing

Querying Clock Tree with get_clock_tree_pins

- Some of the attributes available in get_clock_tree_pins command

Clock QoR:

```
skew_max,  
downstream_delay_max  
downstream_delay_min  
latency_max  
latency_min
```

Topological:

```
height_max  
height_min  
depth_max  
depth_min  
is_reconvergent
```

Pin type/property:

```
is_generated_clock_source  
is_clock_source  
is_sink  
is_leaf  
is_float_pin  
is_explicit_ignore  
is_ignored  
is_cts_added  
is_on_repeater  
is_on_buffer  
is_on_inverter  
is_on_icg  
is_on_macro  
is_inside_etm
```

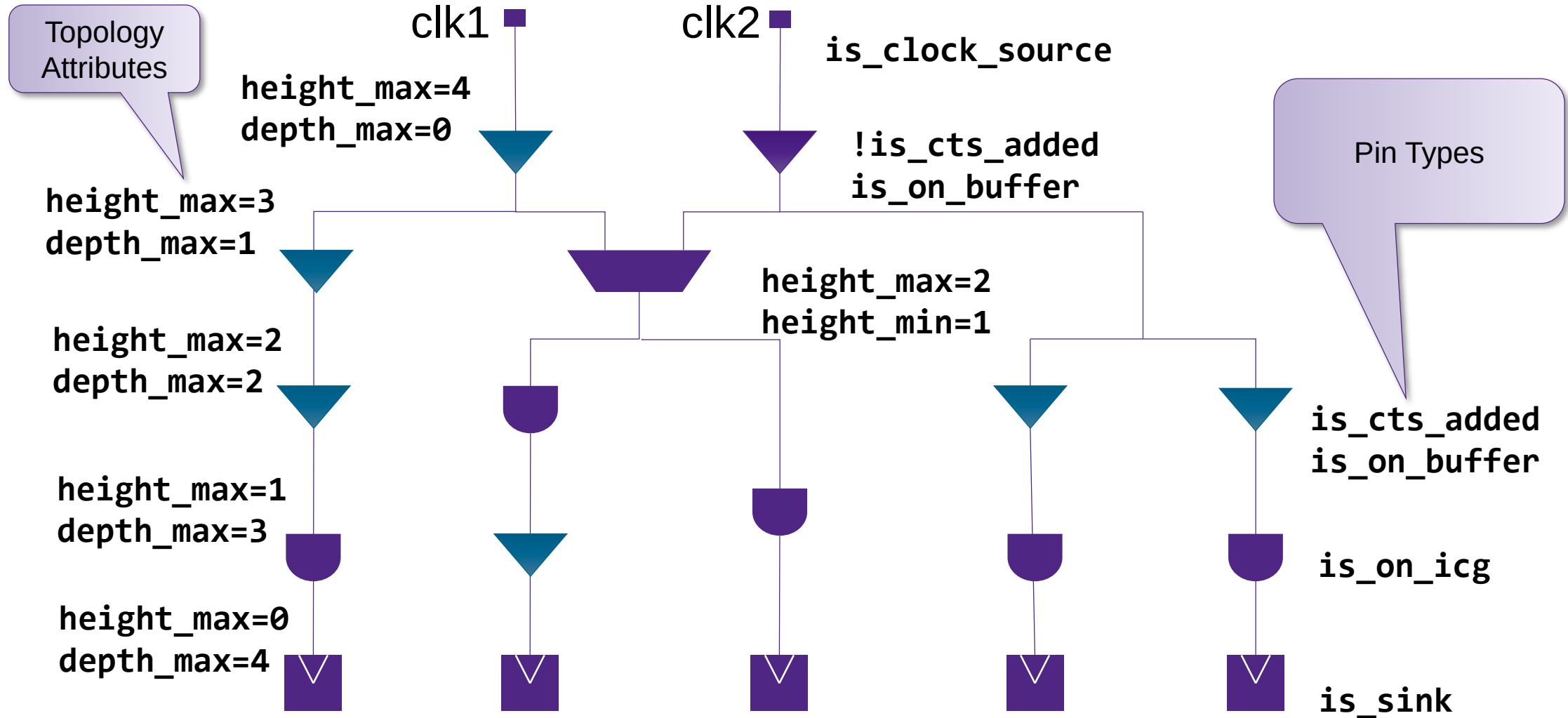
Area/Power related:

```
subtree_num_repeater_max  
subtree_total_area_max  
subtree_total_leakage_power_max  
subtree_total_internal_power_max  
subtree_total_switching_power_max  
cell_internal_power_max  
cell_leakage_power_max  
cell_switching_power_max  
cell_total_power_max  
net_switching_power_max  
toggle_rate_max
```

See the get_clock_tree_pins man page for full list of attributes

Clock Tree Attributes in get_clock_tree_pins

Topology and Pin Type Examples

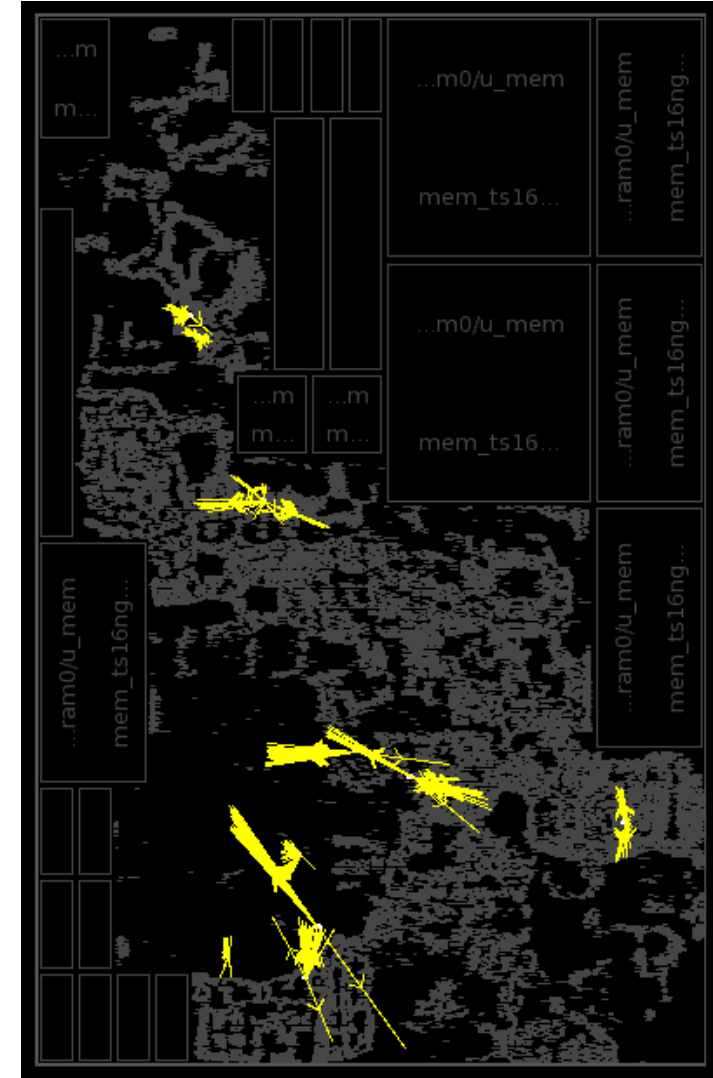


Example - get_clock_tree_pins

- Get_clock_tree_pins can be used to explore any desired parts of any given clock tree, both before or after CTS
- Example:
 - Select 10 highest fanout ICG output pins that drive leaf connections

```
change_selection [get_clock_tree_pins \
  -filter is_on_icg&&is_net_driver&&height_min==0 \
  -sort_by fanout_max -index_range -10]
```

- Highlight net connections
- Overlapping connections may be difficult to see, but could be viewed in a different way



Example - get_clock_tree_pins

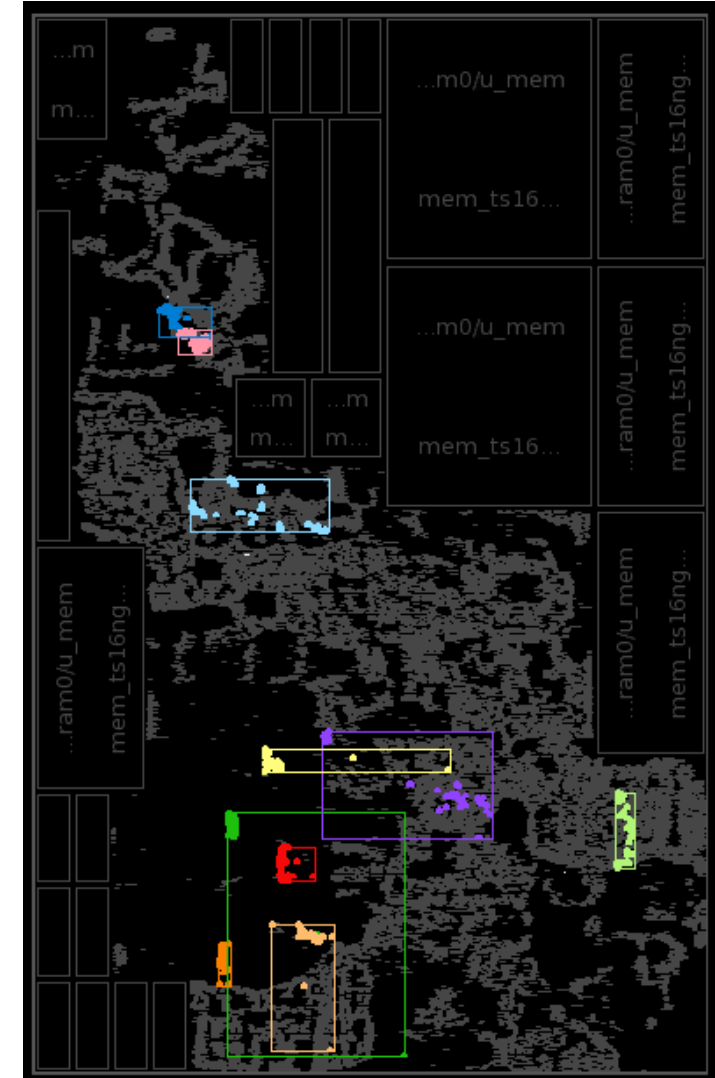
- Get_clock_tree_pins can be used to explore any desired parts of any given clock tree, both before or after CTS
- Example:
 - Select 10 highest fanout ICG output pins that drive leaf connections

```
change_selection [get_clock_tree_pins \
  -filter is_on_icg&&is_net_driver&&height_min==0 \
  -sort_by fanout_max -index_range -10]
```

- Highlight net connections
- Overlapping connections may be difficult to see, but could be viewed in a different way

```
draw_load_bbox [get_selection]
```

[Code for draw_load_bbox in appendix]



get_clock_tree_pins

Example Usage

- Buffers and inverters inserted by CTS

```
icc2_shell> get_flat_cells -of_objects [get_clock_tree_pins \  
-filter is_on_repeater&&is_cts_added]
```
- Get all clock tree pins which are ignored for skew balancing:

```
icc2_shell> get_clock_tree_pins -filter is_ignored
```
- Get sinks inside ETM:

```
icc2_shell> get_clock_tree_pins -filter "is_inside_etm && is_sink"
```
- Get all downstream sinks beyond a pin (traverse through dividers and ICGs):

```
icc2_shell> get_clock_tree_pins -filter is_sink -from mux/Z
```
- Collect and store all latency values into a Tcl array:

```
icc2_shell> get_clock_tree_pins -clocks CLK -sort_by latency_max \  
-assign_to_variable my_array
```

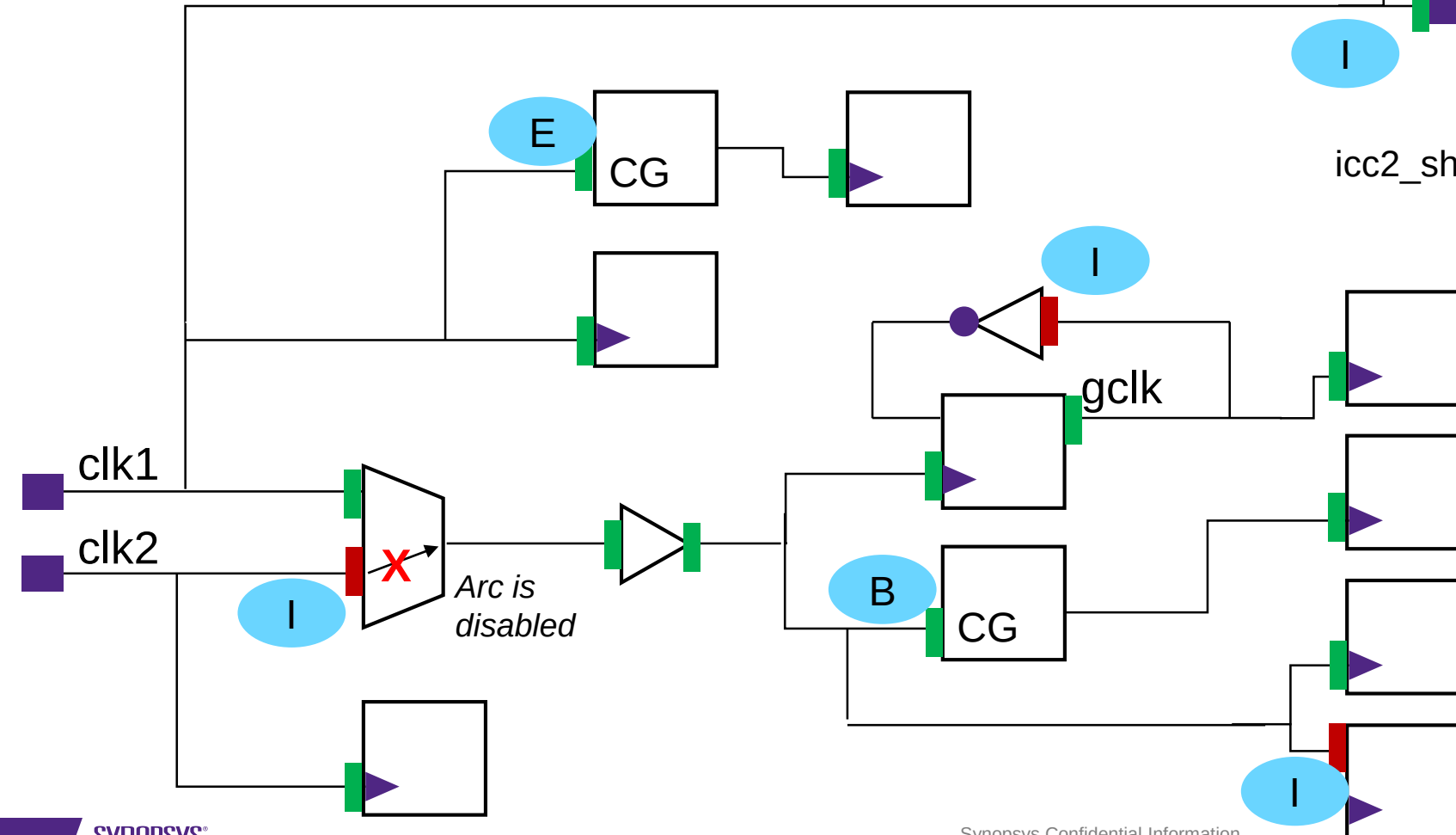
Clock Related Attributes: Pre-CTS

Query pin attributes to debug CTS behavior

```
set_output_delay 2 clk_out2 \
  -clock clk1
```

clk_out2 (unconstrained)

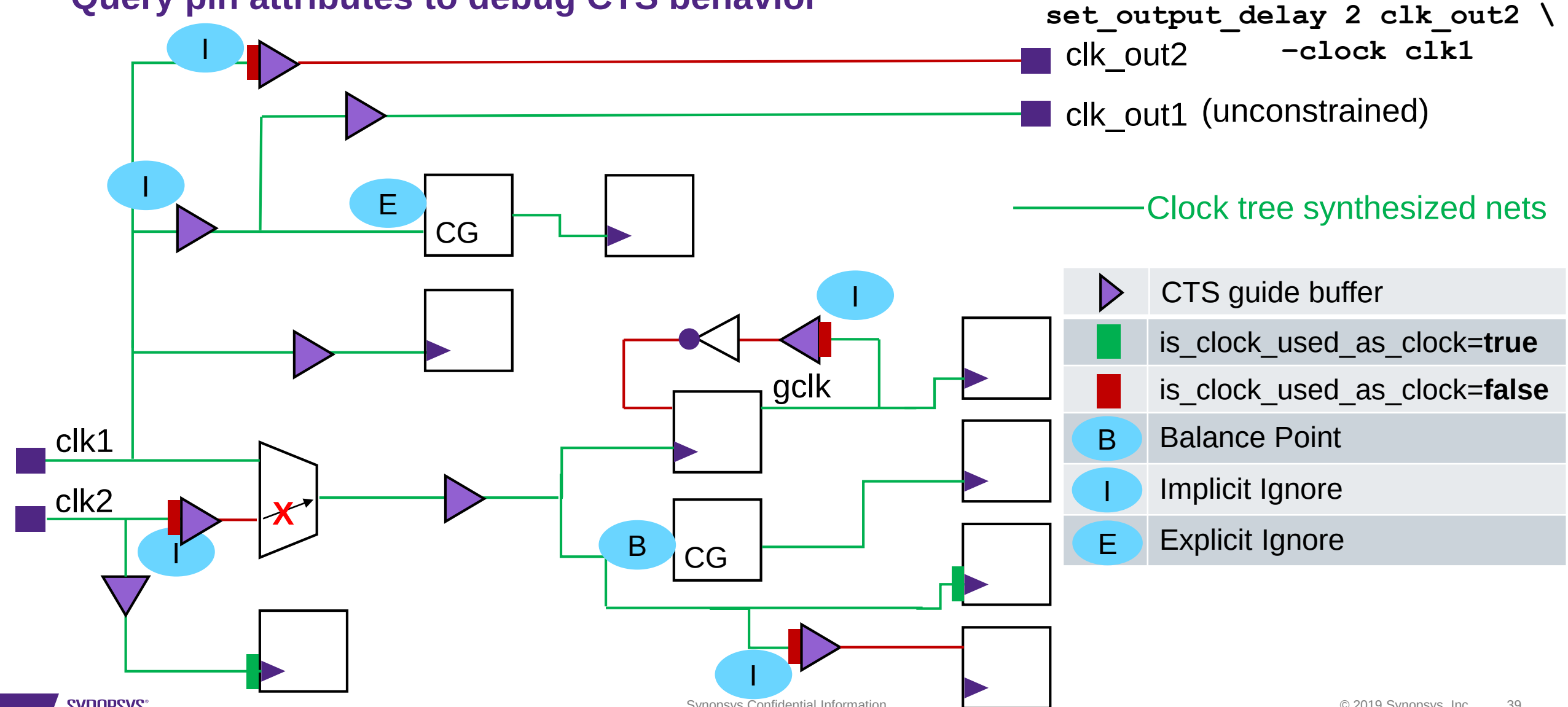
icc2_shell> list_attributes -class pin -application



	is_clock_used_as_clock= true
	is_clock_used_as_clock= false
	Balance Point
	Implicit Ignore
	Explicit Ignore

Clock Related Attributes: Post-CTS

Query pin attributes to debug CTS behavior



Clock Tree Synthesis and Optimization

Handling of Clock Exceptions During Clock Tree Synthesis

- Cell Specific attributes

Exception	Allow Sizing	Allow Movement	Buffer removal
<code>physical_status</code> <code>fixed</code> or <code>locked</code>	No	No	No
<code>physical_status</code> <code>legalize_only</code>	Yes	Only to legalize	No
<code>dont_touch</code>	No	Yes	No
<code>size_only</code>	Yes	Yes	No

Summary

- The “reported” problem is not the “actual” issue
- Debug of your problems will be limited by the “tools” in your “tool box”
 - Proficiency won’t come immediately, but rather through use and experience
- Lot of good information on pin/clock attributes in
 - IC Compiler II Timing Analysis User Guide
 - IC Compiler II Implementation User Guide



Appendix

Clock Tree Synthesis and Optimization

Instance name conventions - SolvNet [2556581](#)

Name	Purpose	Example
cts_inv_<digit> or cts_buf_<digit>	Inserted by CTS clustering	cts_buf_4189301507, cts_inv_7655304984
cts_dlydt_<digit>	Inserted by CTS for delay detour	cts_dlydt_547
cts_trgdly_<digit>	Inserted by CTS for target latency	cts_trgdly_789
_*bip* or *_vip* (v for inverter)	Inserted by CTS to separate ignore pin	clk1_bip4
_*bdp* or *_vdp*	Inserted by CTS to separate clock to data path	sink3_Q_bdp3
_vtd or *_btd*	Inserted by CTS to separate disabled path	clk2_btd1
cto_buf_drc_<digit> or cto_inv_drc_<digit>	Inserted by CTO to fix the DRC	cto_buf_drc_1181
cto_buf_<digit> or cto_inv_<digit>	Inserted by CTO for optimization	cto_buf_1416
cto_dtrdly_<digit>	Inserted by CTO for detour	cto_dtrdly_307365
buf_drc_cln<digit> or inv_drc_cln<digit>	Cloned buffered during DRC fixing	buf_drc_cln306737
cto_buf_cln_<digit> or cto_inv_cln_<digit>	Cloned buffer during CTO	cto_buf_cln_306880
ICDB_<digit>	Cell added during balance clock group	ICDB_1457
CTS_MCSB_<digit>	Cell added by CTS for multi fanout skew balancing	CTS_MCSB_8830
ctosc_inst_<digit>	Inserted by post route CTO for optimization	ctosc_inst_2387
ctosc_drc_inst	Inserted by post route CTO for DRC fixing	ctosc_drc_inst_8523
ccd_setup or ccd_hold	Inserted by CCD for setup/hold optimization	ccd_setup_inst234
clk_drv_r	Created by create_clock_drivers	clk_drv_r_4867
msgts_l	Created by synthesize_multisource_global_clock_trees	msgts_l_377

draw_load_bbox

```
proc draw_load_bbox {icg_pins} {
    set color 0
    gui_set_highlight_options -current_color yellow

    foreach_in_collection pin $icg_pins {
        set downstream_cells [get_cells -of [get_clock_tree_pins -quiet -from $pin]]
        set cells_bbox [get_cell_bbox $downstream_cells]
        set bbox_eff [expr ([sizeof_collection $downstream_cells]**0.5) * [get_area_sum $downstream_cells] /
[get_bbox_area $cells_bbox]]
        puts "[get_attribute -name full_name [get_cells -of $pin]] $color [sizeof_collection $downstream_cells]
[get_bbox_area $cells_bbox] $bbox_eff"
        gui_add_annotation -type rect -color [gui_get_color_value -p highlight [expr $color % 11]] -info_tip "ICG:
[get_attribute -name full_name [get_cells -of $pin]]\nConnections:[sizeof_collection $downstream_cells]\nArea:
[get_bbox_area $cells_bbox]\nbbox_eff:$bbox_eff" $cells_bbox
        set color [expr $color + 1]
        gui_change_highlight -add -collection $downstream_cells
        gui_set_highlight_options -next_color
    }
}
```

get_cell_bbox

```
proc get_cell_bbox {cells} {
    set bounds [list [list [lindex [lsort -real -increasing [get_attribute $cells bbox_llx]] 0] \
        [lindex [lsort -real -increasing [get_attribute $cells bbox_lly]] 0]] \
        [list [lindex [lsort -real -decreasing [get_attribute $cells bbox_urx]] 0] \
        [lindex [lsort -real -decreasing [get_attribute $cells bbox_ury]] 0]]]
    return $bounds
}

proc get_area_sum {cells} {
    set area 0
    foreach_in_collection cell $cells {
        set area [expr $area + [get_attribute -name area -ob $cell]]
    }
    return $area
}
```

Q & A

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